

Yi-Shan (Eason) Chu

RA @ ISS, Academia Sinica | 1st class Math & AI
Graduate | AMIMA

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I am a Research Assistant at the Institute of Statistical Science, Academia Sinica, and graduated from National Chengchi University with a double major in Mathematical Sciences and Computer Science (AI Program), focusing on the mathematical foundations of AI. My research lies at the intersection of theoretical machine learning, learning theory, computational optimization, and geometric perspectives on neural networks. In parallel, I have worked on applied AI problems in remote sensing, computer vision, quantum computing, robotics & UAV systems, and edge-AI systems, with the broader goal of bridging mathematical theory and real-world AI applications.

EDUCATION

B.Sc. in Mathematical Science & Computer Science

Sep. 2023 — Jun. 2026

GPA: 4.15/4.3 (141 credits, Top 10%)

- National Chengchi University - Department of Mathematical Science
- National Chengchi University - Bachelor Program of Artificial Intelligence (Double Major)

RESEARCH EXPERIENCE

Full-time Research Assistant, Academia Sinica - Institute of Statistical Science

Jun. 2026 — Present

Supervised by Dr. Gaspard Bernard

- Study **spectral methods for signal-dimension estimation**, with emphasis on eigenstructure-based testing, threshold calibration, and weak eigenvector identifiability in Gaussian covariance models.

Independent Research Contractor, Taiwan Space Agency (TASA)

Jul. 2024 — Present

- Researched and developed **real-time object detection systems** for **edge-AI computing architectures** in support of upcoming **FORMOSAT-8 satellites**.
- Designed **generative machine learning models** and workflow to reconstruct **optical-band information from Synthetic Aperture Radar (SAR) imagery** with improved efficiency and accuracy.
- Redesigning a **satellite disaster-analysis pipeline** using **Transformers** and **DeepLab-based models** to improve detection consistency, accuracy, and efficiency.

Research Assistant, National Chengchi University - STAR Lab, Dept. of C.S.

Jun. 2025 — Mar. 2026

Supervised by Prof. Jyi-Shane Liu

- Led research on **no-GPS pure visual geo-localization** using only UAV and remote sensing imagery.
- Designed and implemented a **vision-based UAV localization framework** that aligns **UAV observations** with **satellite imagery** in a learned feature space.

Research Assistant, Academia Sinica - Institute of Statistical Science

Sep. 2024 — Mar. 2026

Supervised by Prof. Yuan-chin Ivan Chang

- Conducted research in **computational learning theory**, with focus on **learning capacities of machine learning models** and **model generalization bounds**.
- Led an **NSTC-funded research project** investigating the computational learning properties of **Kolmogorov-Arnold Networks (KANs)**.
- Developed and optimized **active learning algorithms** with **multi-arm bandit strategies**.
- Conducted **large-scale experiments** on **PCA-guided sampling methods** under **high-dimensional settings**.

Research Assistant, National Chengchi University - AI Center

Sep. 2023 — Jun. 2024

- Developed **edge intelligence applications** on NVIDIA Jetson platforms, optimizing real-time inference for **fall-detection systems**.
- Researched **quantum generative models**, including **diffusion-based approaches**, and designed **quantum circuits** for **Chinese calligraphy image generation**, with successful implementation in both **simulation** and **real quantum hardware**.
- Investigated **novel visual descriptors** for **Simultaneous Localization and Mapping (SLAM)** by combining the **Segment Anything Model (SAM)** with classical visual descriptors such as **ORB** and **SIFT**.

RESEARCH INTERESTS

- Primary: Theoretical Machine Learning · Learning Theory · Geometry Perspectives of Neural Networks · Mathematical Image Analysis · Mathematical Foundation of AI
- Secondary: Quantum Machine Learning · Edge Intelligence · Generative AI · Visual-based Geo-localization · Computer Vision · Computational Optimization · Remote Sensing Analysis · Spectral Methods

TECHNICAL & RESEARCH SKILLS

Programming & Scientific Computing

- Python · C/C++ · MATLAB · Julia · Java · R · Kotlin · LaTeX · C#(.NET) · JavaScript
- Linux development · Git · Docker · CUDA programming · Parallel computing · Slurm-based workflows

Machine Learning & AI Systems

- PyTorch · TensorFlow/Keras · scikit-learn · OpenCV · Vitis-AI
- Model training · Optimization · Evaluation · Deployment · Edge Inferences · Agentic AI · Harness Engineering

Quantum Computing & Quantum Machine Learning

- PennyLane · Qiskit · TensorFlow Quantum · IBM Quantum

Systems & Platforms

- NVIDIA Jetson · AMD Xilinx · ROS · Cloud computing platforms · Proxmox VE · VMware · Redis · MySQL

ACADEMIC AWARDS & HONORS

- **Undergraduate Research Grant** - *National Science and Technology Council (NSTC), Taiwan, 2025*
- **Honorable Mention** - *National Taiwan Normal University Mathematics Seminar 2025*
- **First Prize, System-Holdings Student Prize (Best Student Paper)** - *1st IMA AI/ML Congress, UK, 2024*
- **Outstanding Award in Cross-disciplinary AI Applications** - *National Chengchi University, Taiwan, 2024*
- **Honorable Mention** - *SOC-iCaps Smart Computing Cross-Domain Competition, National Cheng Kung University, Taiwan, 2024*
- **Honorable Mention** - *Ephys Challenge Quantum Technology Application Ideas Competition, National Chengchi University, Taiwan, 2024*
- **Finalist** - *2nd Smart Computing Innovative Applications Competition, National Cheng Kung University, Taiwan, 2024*
- **Finalist** - *AI Cross-Domain Sustainable Innovation Competition, National Chengchi University, Taiwan, 2024*
- **Finalist** - *28th National Information Application Service Innovation Competition, Chinese Society of Information Management (CIMS), Taiwan, 2023*

PUBLICATIONS & PREPRINTS

Peer-reviewed Journal Publications

* denotes the corresponding author

1. **Yi-Shan Chu***; Hsuan-Cheng Wei. "Post-Disaster Affected Area Segmentation with a Vision Transformer (ViT)-based EVAP Model using Sentinel-2 and Formosat-5 Imagery." *Terrestrial, Atmospheric and Oceanic Sciences*, in press, 2026. DOI: <https://doi.org/10.1007/s44195-026-00124-1>

Conference Proceedings & Presentations

1. Hsiu-Chuan Hsu*; **Yi-Shan Chu**; Yu-Jung Hsu; Yi-Chieh Wu. "Facilitating Cursive Chinese Calligraphy Generation with Quantum Computing Technology." In *Proceedings of the 2024 10th International Conference on Science and Technology (ICST UGM) & 5th International Conference on Smart and Innovative Agriculture (ICoSIA)*, Universitas Gadjah Mada, 2024. Subsequently published in *Engineering Headway*, Vol. 27 (2025), pp. 49-57. DOI: <https://doi.org/10.4028/p-BZ71fA>
2. **Yi-Shan Chu***; Li-Yu Chang, "Improvement of EVAP for Disaster Analysis Using Transformer-based AI Models," *TAIWAN International Assembly of Space Science, Technology, and Industry (TASTI 2024)*, Taiwan Space Agency, December, 2024.
3. **Yi-Shan Chu***; Ching Wang, "Intention Travel Recommendation System: A Travel Recommendation System Based on Online Comments," 1st Unlocking the Potential: The IMA AI/ML Congress 2024, Institute of Mathematical and its Application, UK, September 2024.

Preprints

1. **Yi-Shan Chu**. "S-GAI: Spectral Geometry-Aware Initialization for Sigmoidal MLPs -- From Dataset Geometry to Network Weights." [arXiv:2606.28444 \[cs.LG\]](https://arxiv.org/abs/2606.28444), 2026. *Submitted for publication. Accepted by ACCV 2026.*
2. **Yi-Shan Chu***; Yueh-Cheng Kuo, "From Universal Approximation Theorem to Tropical Geometry of Multi-Layer Perceptrons." [arXiv:2510.15012 \[stat.ML\]](https://arxiv.org/abs/2510.15012), 2025.

SELECTED PROJECTS

Aliendo Teams, Founder & Team Lead

Oct. 2019 — Present

- Founded and scaled a Minecraft gaming community to over 5,000 registered players, leading a 20-member team across server operations, content design, and community management.
- Engineered, deployed, and maintained multi-server infrastructure, including custom plugin development in Java, Groovy, and Kotlin, VPS deployment, and security hardening.
- Expanded operations into B2B services by providing VPS hosting and technical support to external Minecraft server operators.

An Extensive Study on Computational Learning Capacities in Kolmogorov-Arnold Networks (KAN): A Theoretical and Applied Study, NSTC-funded Project Lead, supervised by Prof. Yuan-Chin Chang

July 2025 - Feb. 2026

- Derived VC-dimension and pseudo-dimension bounds for fixed-knot spline-based KAN models, providing a theoretical foundation for their learnability in the PAC framework.
- Developed Rademacher complexity-based generalization bounds and sample-complexity guarantees to connect capacity analysis with practical learning performance.
- Investigated Kolmogorov-Arnold representation theory and its implications for neural approximation, model capacity, and computational learning theory.